Himanshu Hansaria

Machine Learning Engineer | AI Engineer

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PROFESSIONAL SUMMARY

AI/ML Engineer with 8+ years of experience spanning research, finance, and retail, delivering production-ready AI systems, GenAI agents, and computer vision solutions. Published NeurIPS '23 (spotlight) researcher with hands-on expertise in, LangChain, Databricks and PyTorch, specializing in scalable AI/ML solution development, end-to-end ML pipeline management, and model deployment and monitoring. Proven ability to ship production pipelines—from time-series forecasting at 20K+ SKU scale to RAG-powered agents and multi-GPU self-supervised learning—with measurable impact on accuracy, efficiency, and business decisions.

SKILLS

Languages: Python, JavaScript, SQL, R, C/C++, Java, C#

AI/ML Frameworks: PyTorch, TensorFlow, Keras, Hugging Face Transformers, scikit-learn, XGBoost, LightGBM, LangChain, OpenAI API, OpenCV, NLTK, SpaCy; time-series forecasting (ARIMA, Prophet, PatchTST, Chronos)

Core Competencies: Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Agentic AI, Tool-Calling, Multi-Agent Systems, LLM–API Integration, Prompt Engineering, LLM Evaluation & Observability, AI Guardrails & Security, Fine-tuning of ML and Deep Learning Models, Feature Engineering and Data Preprocessing, Time-Series Data Analysis and Forecasting, Model Performance Tracking and Optimization, Hyperparameter Tuning, Distributed Training, Computer Vision

MLOps & Cloud: Azure Databricks, GCP Vertex AI, MLflow, Apache Flink, CI/CD for ML, Model Deployment and Monitoring, Data Orchestration and Pipeline Automation, End-to-End ML Pipeline Management, PostgreSQL, MySQL, MongoDB

APIs & Backend: FastAPI, Flask, RESTful APIs, Docker, React, React Native

WORK EXPERIENCE

eClerx, New York City Metropolitan Area, NY **Aug 2026 - Present**

AI Engineer

- Architected an end-to-end lead intelligence platform using Python, FastAPI, TypeScript, exchangelib (EWS), and the OpenAI Agents to ingest enterprise mailbox data, reconstruct multi-party conversation threads, and execute parallelized LLM pipelines with structured JSON outputs for summaries, lead scores (0–10) with rationale, and actionable follow-ups, replacing manual inbox review with consistent, automated prioritization of high-intent sales leads.
- Developed an AI-driven leads discovery platform with specialized agents to ingest SEC revenue data, classify thousands of target accounts, retrieve earnings-call transcripts through quota-aware caching, extract operational pain points and business initiatives, and validate signals against verbatim source quotes; built a React dashboard and News-Alerts-compatible exports to surface evidence-backed insights, automate account research, and enable auditable sales prioritization.

Five Below, Philadelphia, PA (Contract) **Jun 2024 – Present**

Machine Learning Engineer

- Architected a production-grade multi-agent inventory intelligence platform on Databricks serving 20,000+ SKUs across 2,000 stores, where a Supervisor Agent orchestrated specialized tool-calling agents across Location Intelligence, Product Intelligence, and Out-of-Stock Detection Genie spaces; autonomously identified future stockout risks, quantified business impact using forecasted sales and inventory metrics, and generated prescriptive recommendations including inventory redistribution, replenishment acceleration, and smarter allocation optimization, surfacing actionable insights through a Streamlit dashboard and helping business stakeholders protect demand and mitigate millions of dollars in potential weekly sales loss.
- Developed a production-grade LLM powered model explainability layer using Meta Llama 3.3 70B for enterprise forecasting systems, combining SHAP-based feature attribution with prompt engineering and controlled generation to provide transparent explanations of XGBoost and ensemble model predictions; implemented guardrails, logging, tracing, and automated evaluation workflows to ensure reliable, governable AI outputs at scale.
- Designed and productionized an enterprise-scale retail demand forecasting platform on Databricks using PySpark, LightGBM, Delta Lake, and Unity Catalog, generating 8-week store-item level forecasts through large-scale feature engineering, distributed model training, and automated prediction pipelines, enabling near real-time demand planning across thousands of products and locations.
- Developed enterprise-scale retail demand forecasting pipelines on Databricks for 2,000+ U.S. stores, generating 30-day store-day demand forecasts; evaluated statistical, machine learning, and foundation forecasting models including ARIMA, Prophet,

XGBoost, PatchTST, and Chronos, leveraging advanced feature engineering, data preprocessing, and hyperparameter tuning to identify XGBoost as the optimal forecasting approach, delivering a 15–20% improvement in forecast accuracy over the incumbent Random Forest production model.

- Productionized an ensemble Voting Regressor combining XGBoost and Random Forest models on Databricks; optimized model weights and hyperparameters through extensive experimentation, achieving an additional 10% reduction in RMSE while improving forecast stability and generalization across geographic regions.
- Engineered an LLM-based Weather Intelligence layer that synthesized aggregated weather-pattern distributions across the enterprise store network into standardized natural-language demand signals, delivering actionable weather insights through Power BI dashboards for business stakeholders
- Developed a comprehensive forecasting evaluation framework using rolling-window backtesting and time-based validation to benchmark multiple forecasting models alongside multiple feature-engineering pipelines, enabling data-driven model selection and continuous forecasting improvements across historical time periods.
- Implemented a production monitoring and model drift detection framework to continuously track forecast accuracy and feature behavior across deployment cycles, enabling proactive retraining, feature refinement, and sustained forecasting reliability at scale.

Indiana University, Bloomington, IN **Feb 2023 – May 2023, Jul 2023 - Jun 2024**

Research Assistant – Machine Learning

- Conducted research on biologically inspired representation learning, pre-training and fine-tuning generative, contrastive, predictive, and masked self-supervised models (including VideoMAE and Vision Transformers) using PyTorch, Hugging Face, and distributed multi-GPU infrastructure; evaluated transfer learning performance across downstream image and video understanding tasks, demonstrating that nature-inspired curriculum learning significantly improves representation quality and accelerates Vision Transformer convergence.
- Co-authored a paper accepted at NeurIPS '23 (spotlight) — top-tier recognition representing the top ~3% of submissions — advancing understanding of biological vision priors in deep learning models. (GitHub)
- Served as Associate Instructor for Elements of AI (Fall '22) and Introduction to AI (Spring '23), mentoring 100+ students on foundational ML and AI concepts.

Google Summer of Code (Apache Software Foundation), USA **Jun 2022 – Sep 2022**

Software Engineer

- Architected and developed a full-stack collaborative notebook platform as a next-generation alternative to JupyterHub, leveraging React, Flask, and MySQL to enable real-time notebook state synchronization, shared kernel execution, and multi-user collaboration; engineered production-grade REST APIs for notebook lifecycle management, runtime sharing, collaboration workflows, and state persistence while preserving isolated execution environments and reproducible compute sessions.
- Designed a custom Slurm job spawner to launch Jupyter Notebooks in Docker containers, enabling scalable HPC-backed interactive compute and data orchestration for research teams. (Medium)

Infostreamz Pvt Ltd, Delhi, India **Jun 2019 – Mar 2021**

Machine Learning Engineer

- Developed and deployed a real-time computer vision system for COVID-19 mask compliance monitoring using OpenCV, TensorFlow, Keras, and JavaScript, achieving 93% classification accuracy and enabling automated alerting through live video-stream inference pipelines.

HSBC, Pune, India **Jun 2016 – Apr 2019**

Software Engineer (Machine Learning)

- Built a time-series forecasting Roster Automation Algorithm and Tool in R that improved Global Help Desk staffing efficiency by 70%, reducing over- and under-staffing across forecast horizons.
- Delivered a Customer Segmentation dashboard (Python, JavaScript) categorizing securities holdings to enable targeted cross-sell and upsell, driving a 40% increase in product uptake.
- Automated KYC processes using Distributed Ledger Technology, Google Cloud Vision API, and Face++, reducing manual review time and improving compliance auditability.

Housing.com, Mumbai, India **May 2014 – Jul 2014**

Image Processing Engineer

- Automated generation of 3D architectural models from 2D floor plans using OpenCV and C++, reducing manual modeling time for real estate listings. (GitHub)

PROJECTS

KrishnaGPT – Vedic RAG Assistant

- Built a production-grade Retrieval-Augmented Generation (RAG) assistant using Python, Flask, ChromaDB, and LLMs, enabling semantic search and source-grounded question answering across the Bhagavad Gita and other Vedic scriptures.
- Engineered an end-to-end document ingestion and vectorization pipeline using Unstructured, recursive text chunking, metadata enrichment, and semantic embeddings, enabling chapter- and verse-level retrieval with accurate source citations.
- Implemented evaluation and observability workflows to monitor retrieval relevance, answer faithfulness, latency, and hallucination rates, improving response quality and system reliability.

NLP-Based Job Matching Skills Parser | GitHub

- Engineered a Flask-based NLP pipeline leveraging OpenAI APIs and MongoDB to extract, normalize, and classify hard and soft skills from unstructured job descriptions, enabling scalable taxonomy-driven candidate-job matching.

Notes Organization AI Agent | GitHub

- Engineered a production-grade Agentic ReAct Agent using LangChain, enabling autonomous reasoning, planning, and workflow execution across enterprise document management tasks
- Implemented dynamic tool orchestration and function-calling to integrate with the Google Docs API and macOS Notes API, allowing the agent to manage note categorization, tagging, and knowledge organization, improving information discoverability and operational efficiency with minimal human intervention

AI-Based Team Preference Alignment | GitHub

- Developed a local search–based optimization engine that assigns students to teams by satisfying teammate preferences and avoidance constraints while minimizing a weighted assignment cost function.

EDUCATION AND CERTIFICATIONS

Indiana University, Bloomington, IN — M.S. Data Science (Aug 2021 – May 2023)
Birla Institute of Technology and Science (BITS Pilani), India — M.Sc. Physics & B.E. Manufacturing (Aug 2011 – May 2016)

ACHIEVEMENTS AND LEADERSHIP

- NeurIPS 2023 Spotlight Paper — represented the top ~3% of accepted submissions at one of AI's most competitive venues.
- Represented India at Robocup Iran Open for Humanoid Robotics, conducting R&D in bipedal locomotion.
- Best Technology Project Awardee — 2 consecutive years (2011–2013) at APOGEE, BITS Pilani.